



Simplified SILCO

One-Shot Semantic Segmentation

Course: Fundamentals of Computer Vision

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1. Task Definition

In our project we work on Few-Shot Semantic Segmentation. Unlike traditional semantic segmentation which requires thousands of labeled examples for fixed classes, our task requires the model to segment an object in the Query image by only one labeled Support example. The aim is for the model to generalize to unseen categories by learning the relationship between the support and query images rather than memorizing class based features.

2. Dataset Description

The model was trained and tested on the Pascal VOC (2007) RGB images dataset. We represented the data by “episodes”, where each episode consists of a Support Image, a Support Mask (defining the target object) and a Query Image with actual mask for measuring the accuracy of predicted mask. The bounding boxes are provided by the dataset and we converted them to binary Pixel-level Masks during the episodic sampling process.

3. Data Pre-processing

- **Resizing:** All images and masks are resized to 224×224 pixels to have the same spatial dimension for convolutional backbone.
- **Normalization:** We normalize images using ImageNet values mean [0.485, 0.456, 0.406] and std [0.229, 0.224, 0.225].
- **Mask Generation:** Create binary masks from the bounding box coordinates, where pixels within the box are 1 and those outside the box are 0.
- **Episodic Sampling:** The class-specific VOCEpisodicDataset class samples images of the same class for the support-query pair, which forces the model to learn the similarity between the images.

4. Network Architecture

The architecture is a simplified version of the SILCO (Spatial-Interactive Learning with Conditional features) framework.

- **Backbone:** The feature extractor is based on the ResNet-50 architecture. Layers 0 and 1 are frozen to preserve general low level features and Layers 2 to 4 are fine-tuned to extract high level semantic information.
- **Similarity Guidance Module (SGM):** This is the “Spatial-Interactive” center. It calculates a dense correlation matrix between each pixel in the query feature map and the foreground pixels of the support feature map. Such point-to-point matching allows the model to learn particular parts of objects regardless of their location.
- **Conditioning & Aggregation** The guidance map from the SGM is concatenated to the original query features. This conditions the final layers on the evidence of similarity.
- **Localization Head:** A series of convolutional layers with Batch Normalization and ReLU activation to process the conditioned features to generate final segmentation logits.

5. Loss Function

The model is trained with a hybrid loss function that guarantees both classification accuracy and structural overlap:

$$\text{Loss} = \lambda_1 \times \text{LBCE} + \lambda_2 \times \text{LDice}$$

- **BCE (Binary Cross-Entropy):** Measures the accuracy of the classification.
- **Dice Loss:** Addresses the class imbalance problem, maximizes the Intersection over Union (IoU) between the prediction and the ground truth.
- **Weights:** $\lambda_1 = 1$ and $\lambda_2 = 1$, o give equal importance to both

6. Hyperparameters

- **Learning Rate:** 1×10^{-5} for Backbone (to avoid large changes of weights in pre-trained layers), 1×10^{-4} for Head and SGM.
- **Batch Size:** 8 , suited for kaggle gpu.
- **Optimizer:** Adam optimizer is chosen for its adaptive learning rate properties which are useful in sparse gradient landscapes encountered in few-shot learning.
- **Scheduler:** ReduceLROnPlateau(factor=0.5, patience=5 epochs)

The values are selected based on the default setting in few-shot segmentation from the original SILCO and PFENet papers, and manually tuned to stabilize the loss curve on the reduced-scale VOC dataset.

7. SOTA Comparison

The model achieves a **Mean IoU of 0.8138 and 96.91% Accuracy**. These results show that the model performs well in the scope of this project. They indicate that the model can localize and segment the target objects with high accuracy. Background pixels inflate the pixel accuracy, so mIoU is a more reliable metric and confirms the strong overlap between the predicted masks and ground-truth regions.

The model performs better than the benchmark, SILCO/PFENet results (approx. 0.53-0.61 mIoU). However, this gap is impacted by the evaluation setting, as the project uses the standard VOC 2007 dataset instead of the more strict PASCAL-5ⁱ unseen-class benchmark.

Such a high performance is mainly due to the Similarity Guidance Module (SGM) which performs pixel-wise correspondence instead of global averaging, thus making the model more robust to the variations of object scale and viewpoint. Conditional Features also lead to more accurate object boundaries by improving mask refinement.

Some of our outputs:



